

Notes: Huang, Linck, Mayer & Parsons (RFS, 2026)
Can Human Capital Explain Income-Based Disparities in Financial Services?

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1 Big picture

- **Question.** Do income-based disparities in the quality of financial services partly arise because financial firms allocate better human capital to higher-income markets?
- **Setting.** The paper studies U.S. mortgage loan officers. Loan officers are the workers who interact with borrowers, help process mortgage applications, and can influence both the quantity and quality of lending.
- **Core data advantage.** The authors build a near-comprehensive panel of over 350,000 mortgage loan officers from NMLS Consumer Access, link them to mortgage originations from CoreLogic, foreclosures from ZTRAX, and local demographics from ACS.
- **Main empirical fact.** Loan officer quality differs substantially and persistently. Loan officers have stable differences in sales volume, misconduct, and bad-loan rates.
- **Main allocation result.** Firms allocate less experienced officers, officers with prior misconduct, officers with worse loan quality, and officers with lower past sales to lower-income branches.
- **Internal-labor-market result.** Firms promote stronger loan officers toward higher-income branches and more strongly punish underperformance in high-income branches.
- **Spillover result.** Loan officers are affected by the prior quality of their same-branch colleagues; poor human capital allocation can therefore contaminate the performance of peers.
- **Outcome result.** Loan officer characteristics predict misconduct, early default, and CFPB complaints. Human capital is therefore not only a labor-market outcome; it affects consumer-facing service quality.
- **Mechanism.** The evidence points mainly to profit-maximizing allocation and career incentives, with less support for explanations based purely on discrimination, regulation, or borrower litigiousness.
- **Economic takeaway.** Disparities in financial services are partly generated inside firms through personnel allocation, promotion, and retention rules.

2 Incremental contribution of the paper

- **From access to quality.** Prior work documents that lower-income neighborhoods have fewer branches or worse observable service outcomes. This paper asks whether a hidden input - worker quality - is one supply-side channel behind those gaps.
- **Human capital as a financial-services input.** The paper imports methods from personnel economics into household finance and banking. It shows that the identity and track record of the loan officer materially affect mortgage outcomes.
- **Spatial allocation inside firms.** The paper does not only show that good and bad workers exist. It shows that firms allocate those workers across neighborhoods in a way that reinforces income-based disparities.
- **Beyond quantity of credit.** Many policies, such as CRA-style approaches, emphasize credit quantity. The paper highlights service quality: misconduct, early defaults, and complaints.
- **New micro-data linkage.** The combination of NMLS employment histories, CoreLogic transaction records, ZTRAX foreclosures, CFPB complaints, and ACS ZIP demographics creates a rare worker-loan-consumer-location panel.
- **Dynamic career channel.** The paper goes beyond static sorting by examining hiring, promotion, separation, and reemployment. This reveals how disparities persist and regenerate over time.
- **Peer spillovers.** By showing that colleagues' prior performance predicts focal loan officer outcomes, the paper adds a multiplier channel: a bad allocation may directly affect customers and indirectly affect coworker performance.

- **Policy contribution.** If low-income borrowers receive lower-quality service because firms staff those areas with weaker human capital, policies that only increase branch presence or loan volume may not improve financial-service quality.

3 Data construction and measurement

3.1 NMLS loan officer panel

- The paper uses NMLS Consumer Access, which records mortgage loan officer licenses and registrations.
- The SAFE Act required mortgage loan originators to be licensed or registered and recorded in the NMLS.
- Each loan officer has a unique NMLS identifier that follows the person across firms, branches, and employment spells.
- The authors use annual historical snapshots from 2012 to 2019 and build a national worker-year panel.
- The panel identifies each officer’s employer, job location, employment history, tenure, and legal/regulatory disciplinary actions.

Interpretation: The unique worker ID is central. It lets the authors separate human capital from branch demand, local regulation, or neighborhood composition because the same worker can be tracked over time and across employers.

3.2 Mortgage origination data

- Mortgage transactions come from CoreLogic and cover nearly all U.S. residential mortgages.
- Since 2014, CoreLogic contains NMLS IDs for loan officers associated with originations.
- The authors aggregate transactions to the loan officer-year level.
- Key productivity variables are total number of originations and total dollar volume of loans.
- Customer-base controls are built by mapping borrowers’ ZIP codes to ACS demographics and averaging across each officer’s borrowers.

Interpretation: These data let the paper measure the output of individual loan officers rather than branch-level lending alone. The customer-base controls help separate officer quality from the borrower pool served by the officer.

3.3 Foreclosure and early-default data

- Foreclosure data come from Zillow’s Transaction and Assessment Dataset (ZTRAX).
- ZTRAX foreclosure records are linked to CoreLogic mortgages using parcel identifiers and county FIPS codes.
- A loan is classified as an early default if it enters foreclosure within twelve months of origination.
- The authors focus on early defaults because they are more closely tied to origination quality than later defaults, which are more affected by macro or household shocks after the loan is made.

Interpretation: Early defaults proxy for bad lending quality. They capture whether the loan officer helped originate loans that failed quickly, which is a more direct measure of poor financial service than long-run default alone.

3.4 Misconduct and complaints

- Misconduct comes from legal and regulatory disciplinary actions attached to the loan officer in NMLS.
- Misconduct is rare, but it captures severe violations that are formally recorded.
- CFPB complaints are measured at the ZIP code-year level for mortgage-related complaints with an available five-digit ZIP code.
- Complaints are scaled by total mortgage originations in the ZIP code-year.

Interpretation: Misconduct is an officer-level measure; complaints are a location-level measure. Their combination lets the paper connect individual human capital to neighborhood service quality.

3.5 Income and customer-base controls

- Local income is measured using ZIP code per capita income from rolling five-year ACS data.
- High income is an indicator for branches in ZIP codes above the sample median income.
- Customer-base controls include average customer-ZIP income, bachelor's degree share, minority share, unemployment rate, and population density.
- These controls are measured using the ZIP codes of borrowers served by a particular officer, not only the ZIP code where the branch is located.

Interpretation: The distinction between branch location and customer base is important. A branch in a low-income ZIP code may serve some borrowers outside that ZIP code; customer controls reduce the concern that results simply reflect borrower mix.

4 Empirical specifications and interpretation

4.1 ZIP-level quality gap

The starting point is a ZIP code-year regression linking local income to measures of financial service quality:

$$Quality_{z,t} = \alpha_t + \gamma_{MSA} + \beta \log(Income_{z,t}) + \varepsilon_{z,t}.$$

Interpretation: A negative coefficient in misconduct, early default, or complaints means that lower-income ZIP codes experience worse financial services quality. This establishes the disparity the rest of the paper tries to explain.

4.2 Persistence in officer performance

For loan officer-year outcomes, the paper estimates regressions of current performance on prior performance while controlling for branch-year fixed effects:

$$Y_{i,b,t} = \alpha_{b,t} + \rho Y_{i,t-1} + X'_{i,t} \Gamma + \varepsilon_{i,b,t}.$$

Interpretation: Branch-year fixed effects compare officers within the same local work environment. Persistence after these fixed effects indicates that performance differences are attached to workers, not only locations.

4.3 Person fixed effects and human-capital importance

The paper compares models with and without loan officer fixed effects:

$$Y_{i,b,t} = \alpha_{b,t} + Controls'_{i,b,t}\Gamma + \eta_i + \varepsilon_{i,b,t}.$$

Interpretation: The increase in adjusted R^2 after adding officer fixed effects measures how much stable worker-level heterogeneity explains service outcomes. Large increases imply that human capital is economically important.

4.4 Empirical Bayes productivity estimates

The empirical Bayes model for loan volume is:

$$\log(LoanVolume)_{i,b,t} = h_b + \eta_i^{RE} + \varepsilon_{i,b,t}.$$

Interpretation: Traditional fixed effects overstate dispersion because some officers have few observations or volatile outcomes. Empirical Bayes shrinkage produces a more credible distribution of worker productivity.

4.5 Hiring, promotion, firing, and reemployment

Hiring and equilibrium allocation regressions relate branch income to officer characteristics:

$$OfficerCharacteristic_{i,f,t} = \alpha_{f,t} + \beta \log(Income_{branch}) + \varepsilon_{i,f,t}.$$

Promotion, firing, and reemployment regressions relate career outcomes to officer quality and location:

$$CareerOutcome_{i,b,t+1} = \alpha_{b,t} + \beta_1 Performance_{i,t} + \beta_2 Performance_{i,t} \times HighIncome_b + Controls'\Gamma + \varepsilon_{i,b,t}.$$

Interpretation: These regressions ask whether firms actively reallocate stronger workers toward richer branches and punish weak performance more strongly in rich branches. Interactions with high-income locations reveal whether career consequences vary by market income.

4.6 Spillover effects

The spillover specification uses coworkers' prior-branch performance:

$$Y_{i,b,t} = \alpha_{f,t} + \alpha_z + \eta_i + \beta \overline{PriorPerformance}_{-i,b,t} + Controls'\Gamma + \varepsilon_{i,b,t}.$$

Interpretation: Using colleagues' performance at their prior assignments helps avoid mechanically attributing local branch conditions to peer effects. The test asks whether the human capital of coworkers changes focal officer outcomes.

5 Identification strategy and threats

5.1 Core identification logic

- The paper's central identification challenge is separating supply-side human capital from local demand and borrower composition.

- It uses within-branch or branch-year comparisons to compare loan officers operating in the same local market at the same time.
- It tracks officers through time to distinguish stable worker differences from temporary branch conditions.
- It controls for customer base demographics to reduce the possibility that low-income branches simply serve riskier borrowers.
- It examines hiring, promotion, and firing decisions to show how observed sorting is produced through firm personnel practices.

5.2 Why branch-year fixed effects matter

Branch-year fixed effects absorb all time-varying branch-level factors: local mortgage demand, branch-level managerial policies, lender strategy, local macro conditions, and local regulation. If two officers work in the same branch-year but have different misconduct or default outcomes, the remaining difference is more likely to reflect officer human capital.

5.3 Why prior performance is useful

The paper frequently uses prior misconduct, prior early default, and lagged loan volume. These measures predate the current location and therefore are less likely to be caused by the current branch's customers. When prior poor performance predicts assignment to low-income branches, the evidence points to selection of workers, not only a contemporaneous treatment effect.

5.4 Threats and responses

- **Borrower risk differences.** Low-income borrowers may have higher default risk. The paper controls for customer demographics and uses early default, which is more linked to origination quality than later default.
- **Local demand differences.** Branch-year fixed effects absorb local demand and branch operations in many specifications.
- **Worker specialization.** Some officers may specialize in particular markets. The paper uses prior performance and within-firm tests to separate specialization from quality.
- **Regulation and CRA incentives.** Mechanism tests examine whether regulatory pressure explains the sorting patterns; evidence favors profit and career incentives.
- **Discrimination.** The paper considers minority share and related controls; the main sorting result is based on income and does not rely solely on race or ethnicity.
- **Descriptive versus causal claims.** Some tests, especially Table 10, are interpreted cautiously because they do not fully distinguish treatment from selection. The paper's strongest causal-style evidence comes from within-branch comparisons and career-transition patterns.

6 Tables and figures

6.1 Table 1: ZIP code income and the quality of financial services

Read:

- The table regresses ZIP code-year service quality on log income, with year and MSA fixed effects.
- Higher income is associated with lower misconduct, lower early default, and fewer CFPB complaints.
- The dependent-variable means are small, so even small coefficients can be meaningful in relative terms.

Interpretation: Table 1 establishes the motivating disparity. It does not yet explain why the disparity exists; it shows that poorer ZIP codes experience lower measured financial-service quality across several outcomes.

Economic message: The rest of the paper asks whether the spatial allocation of human capital helps generate this pattern.

Table 1
ZIP code income and the quality of financial services

	(1)	(2)	(3)
	Misconduct	Early default	CFPB complain
log(income)	−0.0006*** (0.0002)	−0.0024*** (0.0001)	−0.0025*** (0.0001)
Year FE	Y	Y	Y
MSA FE	Y	Y	Y
Observations	146,979	146,979	146,979
R-squared	.0220	.0312	.1161
Dep. var. mean	0.0020	0.0025	0.0025

This table reports OLS regressions examining the effect of local income levels on the quality of financial service. The sample includes all ZIP code-years from 2014 to 2019. In columns 1, 2, and 3 respectively, the dependent variable is the instances of loan officer misconduct, the number of early defaults (foreclosure within a year), and the number of CFPB complaints against mortgage lenders, respectively, each scaled by the number of mortgage originated in the ZIP code that year. The key independent variable is the natural logarithm of per capita income in the ZIP code. The standard errors are clustered at the ZIP code level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 2
Summary statistics

	(1) N	(2) Mean	(3) SD	(4) P25	(5) P50	(6) P75
<i>Loan officer characteristics</i>						
Misconduct	1,110,423	0.0021	0.1637	0	0	0
Early default	1,110,423	0.0020	0.0204	0	0	0
Loan volume (million \$)	1,110,423	6.1694	10.2485	0.3445	1.7326	7.4055
log(loan volume)	1,110,423	14.2357	1.9362	12.7499	14.3651	15.8177
Prior misconduct	755,850	0.0101	0.2275	0	0	0
Prior early default	755,850	0.0063	0.0220	0	0	0
Lag log(loan volume)	755,850	14.4663	1.8414	13.1022	14.6725	15.9320
Tenure	1,110,423	5.9375	6.6280	2	3	7
log(tenure)	1,110,423	1.6112	0.7649	1.0986	1.3863	2.0794
Rookie	1,110,423	0.0398	0.1954	0	0	0
Move up within firm	1,110,423	0.0062	0.0786	0	0	0
Separation	1,110,423	0.1801	0.3842	0	0	0
Move down to lower-income area (conditional on separation)	170,204	0.0700	0.2551	0	0	0
<i>Branch location characteristics</i>						
Income (thousand \$)	1,110,423	38.1142	16.8097	26.5450	34.6460	45.4740
log(income)	1,110,423	10.4666	0.3975	10.1866	10.4530	10.7249
High income	1,110,423	0.5232	0.4995	0	1	1
Fraction of colleagues with prior assignment misconduct	726,448	0.0018	0.0218	0	0	0
Fraction of colleagues with prior assignment early default	726,448	0.0406	0.1202	0	0	0
Fraction of colleagues with prior assignment low loan volume	726,448	0.0871	0.1837	0	0	0.0909
<i>Customer base controls (average at customer-ZIP level)</i>						
log(customer-ZIP income)	1,110,423	10.3878	0.2630	10.2154	10.3746	10.5459
Bachelor's degree share	1,110,423	0.3345	0.1230	0.2490	0.3215	0.4068
Minority share	1,110,423	0.3155	0.1900	0.1703	0.2900	0.4191
Unemployment rate	1,110,423	0.0639	0.0240	0.0479	0.0600	0.0758
log(population density)	1,110,423	6.7817	1.3389	6.0609	6.9734	7.6839

This table presents descriptive statistics for our loan officer-year panel. Columns 1–6 present the sample size (N), mean, standard deviation (SD), 25th percentile (P25), 50th percentile (P50), and 75th percentile (P75).

6.2 Table 2: Summary statistics

Read:

- The panel contains over 1.1 million loan officer-year observations.
- Average tenure is about six years, but the distribution is wide.
- Misconduct and early defaults are rare, yet there is enough variation to study quality.
- Branch and customer demographics vary substantially across officer-years.

Interpretation: Table 2 shows that the setting is large enough to study rare outcomes such as misconduct and early foreclosure while still tracking individual workers over time.

Economic message: The scale of the data is the reason the paper can move from branch-level service disparities to worker-level human capital allocation.

6.3 Figure 1: Persistence in loan officer performance and Table 3: Persistence and person fixed effects in loan officer performance

Read:

- Figure 1 shows that high- and low-performing officers remain separated over future years in sales, early defaults, and misconduct.

- Table 3 panel A confirms persistence: prior misconduct predicts misconduct, prior early default predicts early default, and lagged loan volume strongly predicts current volume.
- Table 3 panel B shows that adding loan officer fixed effects greatly increases explanatory power, especially for misconduct and loan volume.

Interpretation: The figure and table jointly show that loan officer performance is not random. Worker identity matters, and performance differences persist after controlling for branch-year conditions.

Economic message: If worker quality were unimportant, the allocation of workers across neighborhoods would not matter much. These results show that it matters a lot.

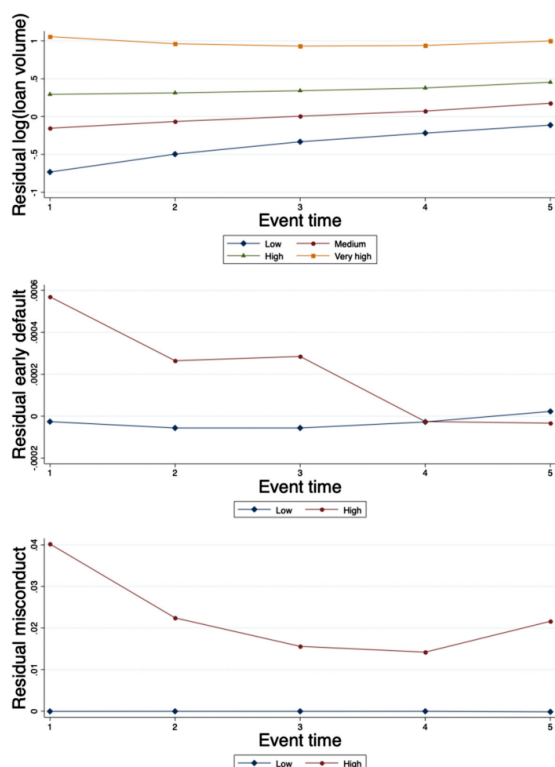


Figure 1
Persistence in loan officer performance

This figure shows the persistence in loan officers' performance in terms of loan volume, loan quality, and misconduct. We regress loan officers' volume/defaults/misconduct on branch-year fixed effects and collect the residuals. For each calendar year, we use the residuals to sort officers into high/low performer portfolios, which we then hold fixed for 5 years, tracking (residual) performance of the officers in event time. We conduct this exercise each calendar year and average portfolio outcomes across event time. In the top plot, we sort loan officer into four groups based on their quartile of residual log(loan volume) in the formation year, and track them for the

(a) Figure 1: persistence in performance

Table 3
Persistence and person fixed effects in loan officer performance

A. Persistence in loan officer performance						
	(1) Misconduct	(2) Early default	(3) log(loan volume)			
Prior misconduct	0.0396*** (0.0112)					
Prior early default		0.0085*** (0.0013)				
Lag log(loan volume)			0.6354*** (0.0016)			
Branch x year FE	Y	Y	Y			
Observations	629,255	629,255	629,255			
R-squared	.2670	.2372	.7355			
Dep. var. mean	0.0009	0.0016	14.7397			
B. Loan officer fixed effects						
	(1) Misconduct	(2)	(3) Early Default	(4)	(5) log(loan volume)	(6)
Customer base controls	Y	Y	Y	Y	Y	Y
Branch x year FE	Y	Y	Y	Y	Y	Y
Loan officer FE	N	Y	N	Y	N	Y
Observations	949,071	837,377	949,071	837,377	949,071	837,377
R-squared	.1508	.5960	.2278	.5130	.5821	.8837
Adjusted R-squared	−.0587	.2437	.0373	.0883	.4790	.7823

This table reports OLS regressions testing whether loan officers' performance is persistent, and assessing the

(b) Table 3: persistence and fixed effects

6.4 Figure 2: Empirical Bayes estimates of loan officers' effect on sales and Figure 3: Loan officer experience, productivity, and lending quality

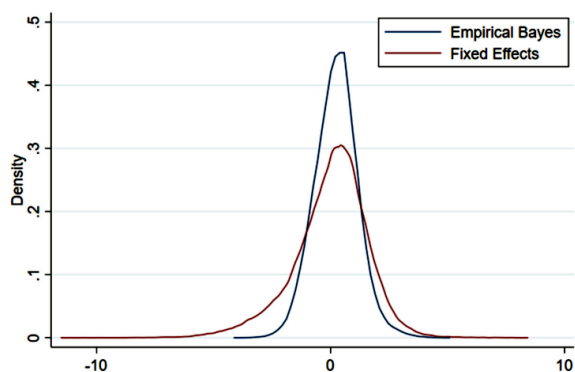
Read:

- Figure 2 shows the distribution of officer effects on loan volume using both traditional fixed effects and empirical Bayes estimates.
- Empirical Bayes shrinks noisy fixed effects, but still shows large dispersion; the 75th percentile officer produces much more than the 25th percentile officer.

- Figure 3 shows that misconduct and early default decline with tenure and with loan volume.

Interpretation: These figures address two concerns. First, productivity dispersion is not only measurement error. Second, high quantity is not mechanically associated with low quality; better officers appear to generate more volume and fewer bad outcomes.

Economic message: The mortgage labor market has meaningful skill differences, and those skill differences combine productivity and quality.



(a) Figure 2: empirical Bayes officer productivity

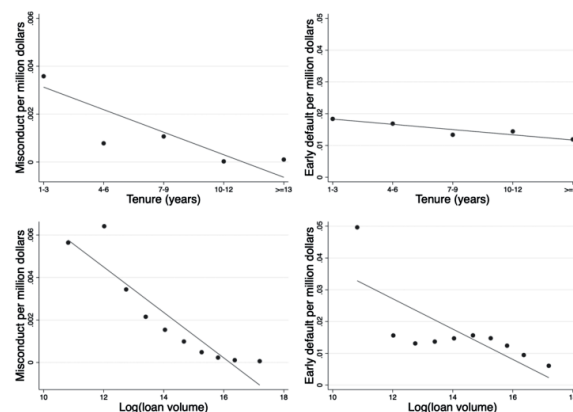


Figure 3
Loan officer experience, productivity, and lending quality
This figure shows how loan officers' experience and lending volume (i.e., productivity), relate to the quality

(b) Figure 3: experience, productivity, and quality

6.5 Table 4: Firms are more likely to hire experienced and skilled loan officers in wealthy areas

Read:

- Rookies are more likely to be hired into lower-income areas.
- Seasoned hires in higher-income areas have lower prior misconduct and fewer prior bad loans.
- Higher-income branches also hire officers with stronger prior loan-volume records.

Interpretation: Table 4 shows that sorting begins at hiring. It is not only that workers develop differently after assignment; firms place workers with different track records into different markets at the start of local employment spells.

Economic message: The initial allocation of human capital favors high-income consumers.

Table 4
Firms are more likely to hire experienced and skilled loan officers in wealthy areas

	(1) Rookie	(2) Prior misconduct	(3) Prior early default	(4) Lag log(loan volume)
log(income)	−0.0463*** (0.0024)	−0.0089** (0.0038)	−0.0025*** (0.0003)	0.3061*** (0.0121)
Firm x year FE	Y	Y	Y	Y
Observations	235,913	141,616	141,616	141,616
R-squared	.1903	.0720	.1114	.3632
Dep. var. mean	0.1788	0.0309	0.0108	14.6209

This table reports OLS regressions examining the effect of ZIP code income levels on the characteristics of the loan officers that firms hire to work at local branches. In column 1, we study all new hires (loan officer-year: where tenure with the firm equals one), and the dependent variable is an indicator for the new hire having no prior experience in the industry (*Rookie*). In columns 2 to 4, we study only seasoned hires (where the officer has prior industry experience), and the dependent variable is *Prior misconduct*, *Prior early default*, and *Lag log(loan volume)*, respectively. The key independent variable is always the natural logarithm of per capita income in the ZIP code where the branch is located. Appendix A defines all variables. The standard errors are clustered at the loan officer level. * $p < .1$; ** $p < .05$; *** $p < .01$.

6.6 Table 5: Firms promote their best loan officers into wealthy areas and Table 6: Firms fire underperforming loan officers, especially in wealthy areas

Read:

- Table 5 shows that tenure and loan volume increase the likelihood of moving up within the firm to a higher-income branch, while misconduct lowers that probability.
- Table 6 shows that misconduct and bad outcomes are punished more strongly in high-income branches.
- Poor sales are also associated with separations, especially in high-income areas.

Interpretation: Together, Tables 5 and 6 show an internal career ladder. Better performance helps workers move toward richer customers; poor performance has harsher consequences in rich branches than in poor branches.

Economic message: Firms appear to set higher personnel standards where customers are more valuable or competition is tougher.

Table 5
Firms promote their best loan officers into wealthy areas

	(1)	(2)	(3)	(4)	(5)
			Move up within firm		
log(tenure)	0.0011*** (0.0001)				0.0009*** (0.0001)
Misconduct		−0.0008** (0.0003)			−0.0006* (0.0003)
Early default			0.0012 (0.0050)		0.0013 (0.0050)
log(loan volume)				0.0008*** (0.0001)	0.0008*** (0.0001)
Customer base controls	Y	Y	Y	Y	Y
Branch x year FE	Y	Y	Y	Y	Y
Observations	949,071	949,071	949,071	949,071	949,071
R-squared	.3475	.3474		.3474	.3476
Dep. var. mean			0.0062		

This table reports OLS regressions examining the effect of loan officers' experience and performance on whether they are promoted within the firm to work in higher-income areas. The sample includes all the loan officer-years in our data from 2014 to 2019. The dependent variable, *Move up within firm*, is an indicator for the loan officer working at one of the firm's other branches that is located in a higher income ZIP code next year. The

(a) Table 5: promotion to wealthy areas

Table 6
Firms fire underperforming loan officers, especially in wealthy areas

	(1)	(2)	(3)	(4)	(5)	(6)
					Separation	
Misconduct	0.0141* (0.0078)	0.0041 (0.0041)				
Misconduct × high income		0.0449*** (0.0079)				
Early default			0.0713*** (0.0225)	0.0260 (0.0304)		
Early default × high income				0.0847* (0.0447)		
log(loan volume)					−0.0248*** (0.0003)	−0.0191*** (0.0004)
log(loan volume) × high income						−0.0102*** (0.0006)
Customer base controls	Y	Y	Y	Y	Y	Y
Branch x year FE	Y	Y	Y	Y	Y	Y
Observations	949,071	949,071	949,071	949,071	949,071	949,071
R-squared	.3953	.3954	.3953	.3953	.4017	.4019
Dep. var. mean				0.1801		

This table reports OLS regressions examining the effect of loan officers' job performance on job separations. The sample includes all the loan officer-years in our data from 2014 to 2019. The dependent variable, *Separation*, is an

(b) Table 6: firing and performance standards

6.7 Table 7: Underperforming loan officers are reemployed in lower-income areas and Table 8: Spillover effects in loan officer performance

Read:

- Table 7 shows that displaced officers with weaker records are more likely to reappear in lower-income areas.
- Less experienced officers are less likely to move down after separation; misconduct and lower volume are associated with downward moves.
- Table 8 shows peer spillovers: colleagues' prior misconduct predicts current misconduct, colleagues' prior early default predicts current early default, and colleagues' prior low volume predicts lower current volume.

Interpretation: Table 7 shows how poor human capital can be reallocated toward poorer neighborhoods across employers, not only within one firm. Table 8 shows why this matters: peer composition affects individual outcomes.

Economic message: Low-income branches can face a double burden: weaker direct assignment of human capital and negative peer spillovers within branches.

Table 7
Underperforming loan officers are reemployed in lower-income areas

	(1)	(2)	(3)	(4)	(5)
		Move down to lower-income area			
log(tenure)	−0.1186*** (0.0014)				−0.1156*** (0.0014)
Misconduct		0.0101** (0.0050)			0.0064 (0.0049)
Early default			−0.0004 (0.0273)		−0.0163 (0.0266)
log(loan volume)				−0.0137*** (0.0005)	−0.0092*** (0.0004)
Customer base controls	Y	Y	Y	Y	Y
Branch FE	Y	Y	Y	Y	Y
Observations	170,204	170,204	170,204	170,204	170,204
R-squared	.3454	.3074	.3074	.3117	.3474
Dep. var. mean			0.0700		

This table reports OLS regressions that examine loan officers' career trajectories following separations from their employer. The sample includes loan officer-years from 2014 to 2019, where the loan officer separate from the firm during the year. The dependent variable, *Move down to lower-income area*, is an indicator for the loan officer being reemployed at another lender's branch located in a lower-income ZIP code next year. The key independent variables are measures of loan officer experience and performance in their current job (pre separation). Customer base controls include average customer ZIP-code-level income, bachelor's degree share

(a) Table 7: reemployment in lower-income areas

Table 8
Spillover effects in loan officer performance

	(1)	(2)	(3)
	Misconduct	Early default	log(loan volume)
Fraction of colleagues with prior assignment misconduct _{std}	0.0010** (0.0004)		
Fraction of colleagues with prior assignment early default _{std}		0.0001*** (0.0000)	
Fraction of colleagues with prior assignment low loan volume _{std}			−0.0339*** (0.0023)
Customer base controls	Y	Y	Y
Firm x year FE	Y	Y	Y
ZIP FE	Y	Y	Y
Loan officer FE	Y	Y	Y
Observations	726,448	726,448	726,448
R-squared	.4209	.4033	.8532
Dep. var. mean	0.0010	0.0017	14.5652

This table reports OLS regressions examining spillover effects in loan officer performance. The sample include all the loan officer-years in our data from 2015 to 2019 (the 2014 data are used to measure officers' prior assignments). The dependent variable is *Misconduct*, *Early default*, and *Log(loan volume)* in columns 1 to 3 respectively. The key independent variables in columns 1 to 3 are the fractions of the focal officer's colleague at the branch with misconduct, early default, and low sales (bottom 10%) at their prior job assignments respectively. We standardize these independent variables by subtracting the mean and dividing by the standard deviation (denoted by the subscript *std*). Customer base controls include average customer ZIP-code-level

(b) Table 8: peer spillovers

6.8 Table 9: The equilibrium allocation of loan officer human capital

Read:

- In the full equilibrium allocation, higher-income branches have fewer rookies, less prior misconduct, fewer prior early defaults, and much higher lagged loan volume.
- Doubling ZIP code income is associated with a lower incidence of rookies and better officer track records.
- The results include firm-by-year fixed effects, so the comparison is within firms in the same year.

Interpretation: Table 9 summarizes the cumulative result of hiring, promotion, firing, and reemployment: better human capital is concentrated in high-income branches.

Economic message: This is the paper's central allocation fact. Financial firms' internal labor markets sort talent toward richer neighborhoods.

6.9 Table 10: The impact of human capital on the quality of financial services

Read:

- Panel A links officer characteristics to misconduct, early default, and CFPB complaints.

Table 9
The equilibrium allocation of loan officer human capital

	(1) Rookie	(2) Prior misconduct	(3) Prior early default	(4) Lag log(loan volume)
log(income)	−0.0059*** (0.0006)	−0.0019* (0.0010)	−0.0014*** (0.0001)	0.4526*** (0.0080)
Firm x year FE	Y	Y	Y	Y
Observations	1,110,423	755,850	755,850	755,850
R-squared	.0777	.0473	.1171	.4364
Dep. var. mean	0.0398	0.0101	0.0063	14.4663

This table reports OLS regressions examining the effect of ZIP code income levels on the characteristics of the loan officers working at local branches. The sample includes all the loan officer-years in our data from 2014 to 2019 (columns 2–4 require at least one prior year of data on officers). In columns 1 to 4, the dependent variable is *Rookie*, *Prior misconduct*, *Prior early default*, and *Lag log(loan volume)*, respectively. The key independent variable is always the natural logarithm of per capita income in the ZIP code where the branch is located. Appendix A defines all variables. The standard errors are clustered at the loan officer level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 10
The impact of human capital on the quality of financial services
A. Quality of financial services

	(1) Misconduct	(2) Early default	(3) CFPB complaints
log(tenure)	−0.0007*** (0.0002)	−0.0001*** (0.0000)	−0.0004*** (0.0001)
Prior misconduct	0.0260*** (0.0083)	0.0001** (0.0001)	0.0000*** (0.0000)
Prior early default	−0.0006 (0.0044)	0.0056*** (0.0008)	0.0065*** (0.0018)
Lag log(loan volume)	0.0001 (0.0001)	−0.0000*** (0.0000)	−0.0003*** (0.0001)
Customer base controls	Y		Y
Branch x year FE	Y		
Loan characteristics controls		Y	
Firm FE		Y	
ZIP x year FE		Y	
Year FE			Y
MSA FE			Y
Observations	629,255	27,811,234	118,521
R-squared	.2649	.0120	.0996
Dep. var. mean	0.0010	0.0018	0.0016

B. Variance decomposition

	(1) Misconduct	(2) Early default	(3) CFPB complaints
log(income)	0.02%	61.19%	73.64%
log(tenure)	0.41%	16.82%	19.88%
Prior misconduct	99.54%	0.29%	0.03%
Prior early default	0.00%	21.20%	6.46%
Lag log(loan volume)	0.02%	0.50%	0.05%
Observations	629,255	27,811,234	118,521

This table reports OLS regressions examining the effect of loan officer characteristics on the quality of financial services provided to borrowers. Panel A focuses on the OLS estimates. In column 1, the unit of observation is

- More tenure generally predicts better outcomes, while prior misconduct and prior early defaults predict worse outcomes.
- Panel B decomposes explanatory power. Officer-related human capital explains a meaningful share of quality variation, especially severe misconduct and CFPB complaints.

Interpretation: Table 10 is the bridge from labor allocation to consumer outcomes. It shows that the worker characteristics sorted across neighborhoods are not merely internal HR variables; they predict the quality of financial services borrowers receive.

Economic message: Human capital is a mechanism through which neighborhood income becomes linked to service quality.

Table 10
The impact of human capital on the quality of financial services

A. Quality of financial services

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log(tenure)	−0.0007*** (0.0002)	−0.0001*** (0.0000)	−0.0004*** (0.0001)
Prior misconduct	0.0260*** (0.0083)	0.0001** (0.0001)	0.0000*** (0.0000)
Prior early default	−0.0006 (0.0044)	0.0056*** (0.0008)	0.0065*** (0.0018)
Lag log(loan volume)	0.0001 (0.0001)	−0.0000*** (0.0000)	−0.0003*** (0.0001)
Customer base controls	Y		Y
Branch x year FE	Y		
Loan characteristics controls		Y	
Firm FE		Y	
ZIP x year FE		Y	
Year FE			Y
MSA FE			Y
Observations	629,255	27,811,234	118,521
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7 Interpretive synthesis

- **Human capital is a production input.** Loan officers differ in sales, misconduct, and loan quality. These differences are durable and economically important.
- **Firms sort this input spatially.** High-income areas receive more experienced and better-performing officers, while low-income areas receive more rookies and officers with poorer records.
- **Sorting is dynamic.** Hiring, promotion, firing, and reemployment all contribute to the equilibrium allocation.
- **Discipline is asymmetric.** Poor performance is punished more in high-income branches. This means low-income branches can become places where underperformance is more tolerated.
- **Peer effects amplify sorting.** Weak human capital at a branch may worsen not only direct outcomes but also coworkers' outcomes.
- **Policy implication.** Improving access to finance requires more than increasing branch or loan quantity; the quality of personnel assigned to underserved neighborhoods matters.

8 Short summary

- Low-income ZIP codes have more loan officer misconduct, more early defaults, and more CFPB mortgage complaints.
- The paper builds a national loan officer-year panel using NMLS, CoreLogic, ZTRAX, CFPB, and ACS data.
- Loan officers differ substantially in productivity and quality, and these differences persist over time.
- Officer fixed effects explain large shares of misconduct, defaults, and sales volume.
- Firms hire more rookies and worse-track-record officers into low-income areas.
- Firms promote experienced, high-volume, lower-misconduct officers toward higher-income branches.
- Firms punish misconduct and poor performance more strongly in high-income branches.
- Displaced underperformers are more likely to be reemployed in lower-income areas.
- Coworker quality affects focal officer outcomes, creating spillovers.
- Human capital characteristics predict service quality and explain a meaningful part of income-based disparities.

9 Conclusion

9.1 Core conclusion

The paper shows that income-based disparities in financial services are partly produced through firms' allocation of human capital. Low-income neighborhoods do not merely have different borrower demand; they are served by different workers with different experience, productivity, and track records.

9.2 Economic intuition

Loan officers are not interchangeable. Some are more productive, avoid misconduct, and originate loans that perform better. Firms have incentives to assign scarce high-quality workers where revenues, competition, and customer value are highest. This sorting leaves low-income consumers with weaker personnel and potentially worse financial advice, worse loan processing, and more harmful outcomes.

9.3 What the paper establishes

The paper establishes three links: worker quality matters; firms sort workers by local income; and sorted worker characteristics predict service quality. The combination makes human capital a plausible and important supply-side channel behind financial-service inequality.

9.4 What the paper does not fully establish

Some outcomes remain descriptive. In particular, the paper cannot perfectly separate selection from treatment in every test, and it cannot observe all dimensions of borrower risk, branch management, or the firm's profit function. The mechanism tests favor profit and career incentives, but the paper does not claim to fully solve the firm's assignment problem.

9.5 Final takeaway

The central message is that financial inequality can be created inside organizations. If policy focuses only on where firms lend or open branches, it may miss a key margin: who serves the customers in those branches.